

Artificial Intelligence Project 4

Image and Text Recognition System using OCR

Abstract

Optical Character Recognition (OCR) is a technology that enables computers to recognize and extract text from images. This project presents an OCR-based image and text recognition system developed using Python, OpenCV, and Tesseract OCR. The system processes input images, applies preprocessing techniques, extracts text, and displays the recognized results. It supports printed text and simple handwritten text recognition.

Introduction

Artificial Intelligence has significantly improved the ability of machines to interpret visual information. One important application of AI is Optical Character Recognition (OCR), which converts text contained in images into machine-readable text. OCR systems are widely used in document digitization, automated data entry, and information retrieval.

This project implements an OCR-based recognition system capable of detecting and extracting text from images.

Problem Statement

Manually typing text from images is time-consuming and prone to errors. The objective of this project is to develop a system that automatically extracts textual information from images and converts it into editable digital text.

Objectives

- Develop an image and text recognition system.
- Extract text from images automatically.
- Apply image preprocessing techniques.
- Generate confidence scores for recognized text.
- Display detected text regions using bounding boxes.
- Save extracted text into output files.

Tools and Technologies

Programming Language

- Python

Libraries

- OpenCV
- PyTesseract

- NumPy
- Pillow

OCR Engine

- Tesseract OCR

Methodology

Step 1: Image Acquisition

The system reads image files from the images folder.

Step 2: Preprocessing

The following preprocessing operations are applied:

- Image resizing
- Grayscale conversion
- Noise reduction
- Thresholding

Step 3: OCR Processing

The preprocessed image is passed to the Tesseract OCR engine which extracts textual information.

Step 4: Confidence Analysis

Confidence scores are generated for each detected word.

Step 5: Text Detection

Bounding boxes are drawn around recognized text regions.

Step 6: Output Generation

The extracted text is stored in text files and processed images are saved.

System Flow

Input Image

↓

Preprocessing

↓

OCR Engine

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Text Recognition

↓

Confidence Analysis

↓

Output Generation

Observations

- Printed text achieved high recognition accuracy.
- Decorative fonts produced moderate accuracy.
- Handwritten text produced lower accuracy due to OCR limitations.

Advantages

- Easy to use
- Fast text extraction
- Supports multiple image formats
- Automatic result storage
- Useful for digitization tasks

Limitations

- Reduced accuracy for complex handwriting.
- Sensitive to image quality.
- Decorative fonts may affect performance.

Future Work

- Integration of deep learning OCR models.
- Support for multiple languages.
- Real-time camera-based recognition.
- Improved handwriting recognition using neural networks.

Conclusion

This project successfully implements an OCR-based image and text recognition system using Python and Tesseract OCR. The system can recognize printed text with high accuracy and provides reasonable performance for simple handwritten text. The project demonstrates the practical application of Artificial Intelligence in text recognition and document digitization.

References

1. Tesseract OCR Documentation
2. OpenCV Documentation
3. Python Official Documentation
4. Optical Character Recognition Research Papers

OUTPUT

```
PS C:\Users\user\Desktop\project#4> python main.py
```

```
=====
Processing: sample1.jpg
=====
```

```
Recognized Text:
```

```
-----
```

```
Jane Doe
```

```
123 Main Street
Besttown, CA 12345
(123) 456-7899
jane.doe@email.com
```

```
Product Manager
Acme Corporation
```

```
Confidence Scores:
```

```
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```

```
Word: Jane | Confidence: 95.0
Word: Doe | Confidence: 96.0
Word: 123 | Confidence: 96.0
Word: Main | Confidence: 96.0
Word: Street | Confidence: 96.0
Word: Besttown, | Confidence: 91.0
Word: CA | Confidence: 96.0
Word: 12345 | Confidence: 96.0
Word: (123) | Confidence: 94.0
Word: 456-7899 | Confidence: 71.0
Word: jane.doe@email.com | Confidence: 88.0
Word: Product | Confidence: 95.0
```

Saved:
output\sample1_detected.jpg

=====
Processing: sample2.jpg
=====

Recognized Text:

SPECTERS OF
OBSIDIAN
TWILIGHT:

A VEIL
BETWEEN REALMS

Confidence Scores:

Word: SPECTERS | Confidence: 92.0
Word: OF | Confidence: 96.0
Word: OBSIDIAN | Confidence: 96.0
Word: TWILIGHT: | Confidence: 95.0
Word: A | Confidence: 96.0
Word: VEIL | Confidence: 95.0
Word: BETWEEN | Confidence: 96.0
Word: REALMS | Confidence: 96.0

Saved:
output\sample2_detected.jpg

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